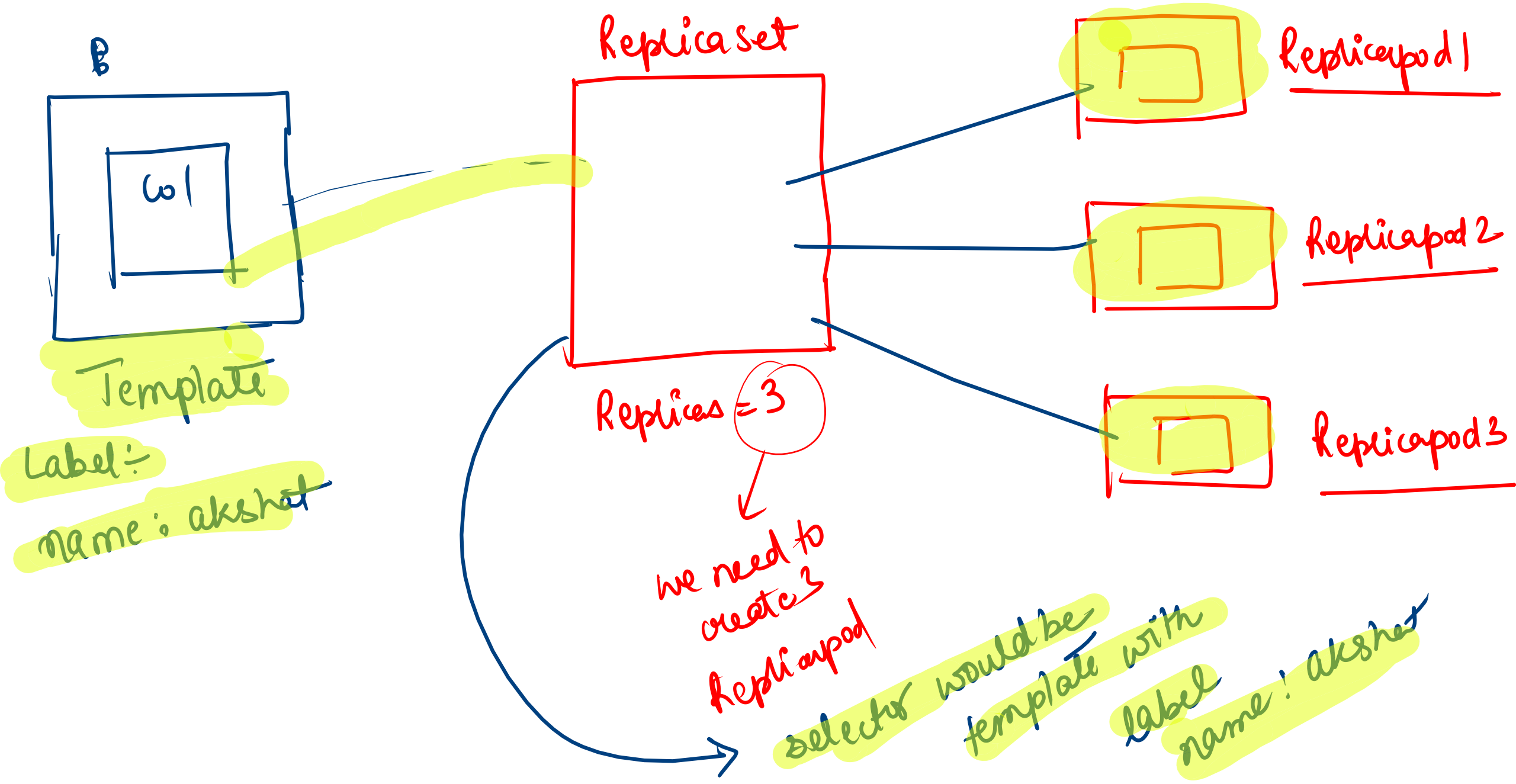
Controller which ensures that the desired number of pods matches the current pod which means that if a pod is deleted due any reason, Replica set will create a Replica pod for you. ← Replica set


Replicaset
Replicapod1
Replicapod2
Replicapod3
Template
Label:-
name: akshat
Replicas = 3
we need to create 3 Replicapod
selector would be template with label name: akshat

vi myrs.yml

```yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: myreplica
spec:
  replicas: 5
  selector:
    matchLabels:
      company: staragile
  template:
    metadata:
      labels:
        company: staragile
    spec:
      containers:
        - name: c01
          image: httpd
          ports:
            - containerPort: 80
```

for the kind as Replicaset the apiversion is apps/v1

The name of replicaset is myreplica

we will create 5 Replica pods

Replica pods will be created from the Template with label company: staragile

The label of the template is company: staragile

Template is telling that containers which will be present in replica pod will have name as c01

Here we are create template

```
root@ip-172-31-6-141:/home/ubuntu# kubectl apply -f myrs.yml
replicaset.apps/myreplica created
root@ip-172-31-6-141:/home/ubuntu#
```

```
root@ip-172-31-6-141:/home/ubuntu# kubectl get pods -o wide
NAME              READY   STATUS    RESTARTS   AGE   IP                NODE
myfirstpod        1/1     Running   0          64m   192.168.9.129     ip-172-31-2-109
mypod             1/1     Running   0          25m   192.168.122.65    ip-172-31-15-78
myreplica-2tl6d   1/1     Running   0          18s   192.168.9.131     ip-172-31-2-109
myreplica-7k9tf   1/1     Running   0          18s   192.168.122.66    ip-172-31-15-78
myreplica-fz4pd   1/1     Running   0          18s   192.168.9.132     ip-172-31-2-109
myreplica-jg2qv   1/1     Running   0          18s   192.168.122.67    ip-172-31-15-78
myreplica-k7krl   1/1     Running   0          18s   192.168.9.130     ip-172-31-2-109
```

5 pods

if u delete the pod it will be replaced with another pod

```
root@ip-172-31-6-141:/home/ubuntu# kubectl get pods -o wide
NAME              READY   STATUS    RESTARTS   AGE   IP                NODE               NOMINATED
myfirstpod        1/1     Running   0          64m   192.168.9.129     ip-172-31-2-109    <none>
mypod             1/1     Running   0          25m   192.168.122.65    ip-172-31-15-78    <none>
myreplica-2tl6d   1/1     Running   0          55s   192.168.9.131     ip-172-31-2-109    <none>
myreplica-7k9tf   1/1     Running   0          55s   192.168.122.66    ip-172-31-15-78    <none>
myreplica-fz4pd   1/1     Running   0          55s   192.168.9.132     ip-172-31-2-109    <none>
myreplica-jg2qv   1/1     Running   0          55s   192.168.122.67    ip-172-31-15-78    <none>
myreplica-k7krl   1/1     Running   0          55s   192.168.9.130     ip-172-31-2-109    <none>
root@ip-172-31-6-141:/home/ubuntu# kubectl delete pod  myreplica-2tl6d
pod "myreplica-2tl6d" deleted
root@ip-172-31-6-141:/home/ubuntu# kubectl get pods -o wide
NAME              READY   STATUS              RESTARTS   AGE   IP                NODE
myfirstpod        1/1     Running             0          64m   192.168.9.129     ip-172-31-2-109
mypod             1/1     Running             0          26m   192.168.122.65    ip-172-31-15-78
myreplica-7k9tf   1/1     Running             0          64s   192.168.122.66    ip-172-31-15-78
myreplica-fz4pd   1/1     Running             0          64s   192.168.9.132     ip-172-31-2-109
myreplica-jg2qv   1/1     Running             0          64s   192.168.122.67    ip-172-31-15-78
myreplica-k7krl   1/1     Running             0          64s   192.168.9.130     ip-172-31-2-109
myreplica-smxss   0/1     ContainerCreating   0          4s    <none>            ip-172-31-15-78
root@ip-172-31-6-141:/home/ubuntu#
```

new pod takes place of deleted pod

get list of replicaset

```
root@ip-172-31-6-141:/home/ubuntu# kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
myreplica     5         5         5       2m31s
root@ip-172-31-6-141:/home/ubuntu# kubectl get replicaset
NAME          DESIRED   CURRENT   READY   AGE
myreplica     5         5         5       2m36s
root@ip-172-31-6-141:/home/ubuntu#
```

Both are same commands

if u delete the replicaset the replicapod will automatically be deleted

```
root@ip-172-31-6-141:/home/ubuntu# kubectl delete rs myreplica
replicaset.apps "myreplica" deleted
root@ip-172-31-6-141:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
myfirstpod    1/1     Running   0          68m
mypod         1/1     Running   0          29m
root@ip-172-31-6-141:/home/ubuntu#
```

Replica pods are deleted

scale number of pods

```
root@ip-172-31-6-141:/home/ubuntu# kubectl apply -f myrs.yml
replicaset.apps/myreplica created
root@ip-172-31-6-141:/home/ubuntu# kubectl scale --replicas=10 -f myrs.yml
replicaset.apps/myreplica scaled
root@ip-172-31-6-141:/home/ubuntu# kubectl get pods
NAME                READY   STATUS              RESTARTS   AGE
myfirstpod          1/1     Running             0          70m
mypod               1/1     Running             0          31m
myreplica-c24x7     0/1     ContainerCreating   0          3s
myreplica-c7jkf     1/1     Running             0          6s
myreplica-cpbzq     1/1     Running             0          6s
myreplica-ctxrb     0/1     ContainerCreating   0          3s
myreplica-fvww2     0/1     ContainerCreating   0          3s
myreplica-jhlq5     1/1     Running             0          6s
myreplica-m97lx     0/1     ContainerCreating   0          6s
myreplica-qvqzb     0/1     ContainerCreating   0          3s
myreplica-v6pbg     0/1     ContainerCreating   0          3s
myreplica-xpv5w     1/1     Running             0          6s
root@ip-172-31-6-141:/home/ubuntu#
```

you can scale or descale the pods